Assessment - Report

Gender Recognition by Voice Analysis

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Credit Member Statement – CP3403

Member Name	Contributions (Percentage)
Lim Dong Xian	Total 50% - Data Description, Data Preprocessing, ARM, Naïve Bayes, Discussion and Comparisons (50%), Future Improvements
Richo Winson Tandoto	Total 50% - Abstract, Introduction, Business Scenario, Features, Decision Tree, Discussion and Comparisons (50%), Conclusion

Link to Gender Recognition by Voice database:
https://www.kaggle.com/datasets/murtadhanajim/vocal-gender-features

All results and images in this document will be on version 20 of our work.

Link to our work name (Assessment 1 v2):
https://www.kaggle.com/code/limdx006/assessment-1-v2

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Abstract

This project examines gender voice recognition technology, which is essential for applications such as speech-based authentication and personalised user experiences. The study addresses challenges such as speech variation and accents by analysing 44 variables to reliably identify a speaker's gender from their voice. The technology has potential in healthcare, security, and consumer electronics, enhancing personalization, targeted marketing, and accessibility. The dataset sourced from Kaggle has been pre-processed and cleaned to facilitate analysis and aims to improve gender voice recognition technology and overcome its limitations.

Introduction

Voice recognition is a technology that allows machines to process and interpret human speech, allowing them to interpret spoken commands (TechTarget, 2024). Voice recognition applies to various scenarios, making it crucial to determine whether a voice belongs to a male or a female. Most machines achieve this by distinguishing different acoustic properties of speech, such as tonal quality and pitch.

Gender voice recognition plays a crucial role in areas such as speech-based authentication and linguistic research. However, several factors, such as speech variation and accents, can hinder accurate gender classification. To address these challenges, we analysed 44 different variables to assess whether a speaker's gender can be reliably identified from their voice. This study aims to contribute to the advancement of gender voice recognition technology and overcome its limitations.

Business Scenario

Gender voice recognition technology is now more commonly integrated across various industries, including healthcare, security, and consumer electronics. In healthcare, it supports personalised speech therapy and diagnostic tools by customizing responses based on a speaker's vocal characteristics. Security agencies also benefit from gender identification by using vocal cues to assist in investigations where only audio evidence is available. Meanwhile, technology companies can use this feature to create more personalized experiences in voice assistants and smart devices to enhance user satisfaction and engagement.

Beyond personalization, gender voice recognition can also refine targeted marketing strategies based on vocal analytics and improve accessibility features and user interaction design. Gender voice recognition can also help businesses understand their customers better and deliver responses that satisfy the customer, which can make systems smarter and more inclusive.

Data Description

Gender Recognition by Voice dataset is obtained from Kaggle, it contains 16,148 rows and 44 columns. It consists of various vocal features extracted from audio recordings, which are used to classify gender (0 for Female, 1 for Male). The data set seems to have done some of the data cleaning and processing, which ensures a good starting point for analysing gender voice recognition. It also has a usability rate of 10.0, which may suggest that a lot of people do find this dataset useful.

Data info

Using the info function, we can see that this database contains 43 float64 and one int64 data type variable.

```
RangeIndex: 16148 entries, 0 to 16147
Data columns (total 44 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   mean_spectral_centroid  16148 non-null  float64
 1   std_spectral_centroid   16148 non-null  float64
 41  mfcc_13_mean            16148 non-null  float64
 42  mfcc_13_std             16148 non-null  float64
 43  label                   16148 non-null  int64
dtypes: float64(43), int64(1)
memory usage: 5.4 MB
```

Figure 1 & 2: output data.info()

Correlation Heatmap

A correlation heatmap helps in identifying relationships between variables. Based on Figure 3, most features show low to moderate correlation, meaning many variables are not strongly connected. However, a few features like mean_spectral_centroid and std_spectral_centroid show strong positive correlation, which may indicate some redundancy.

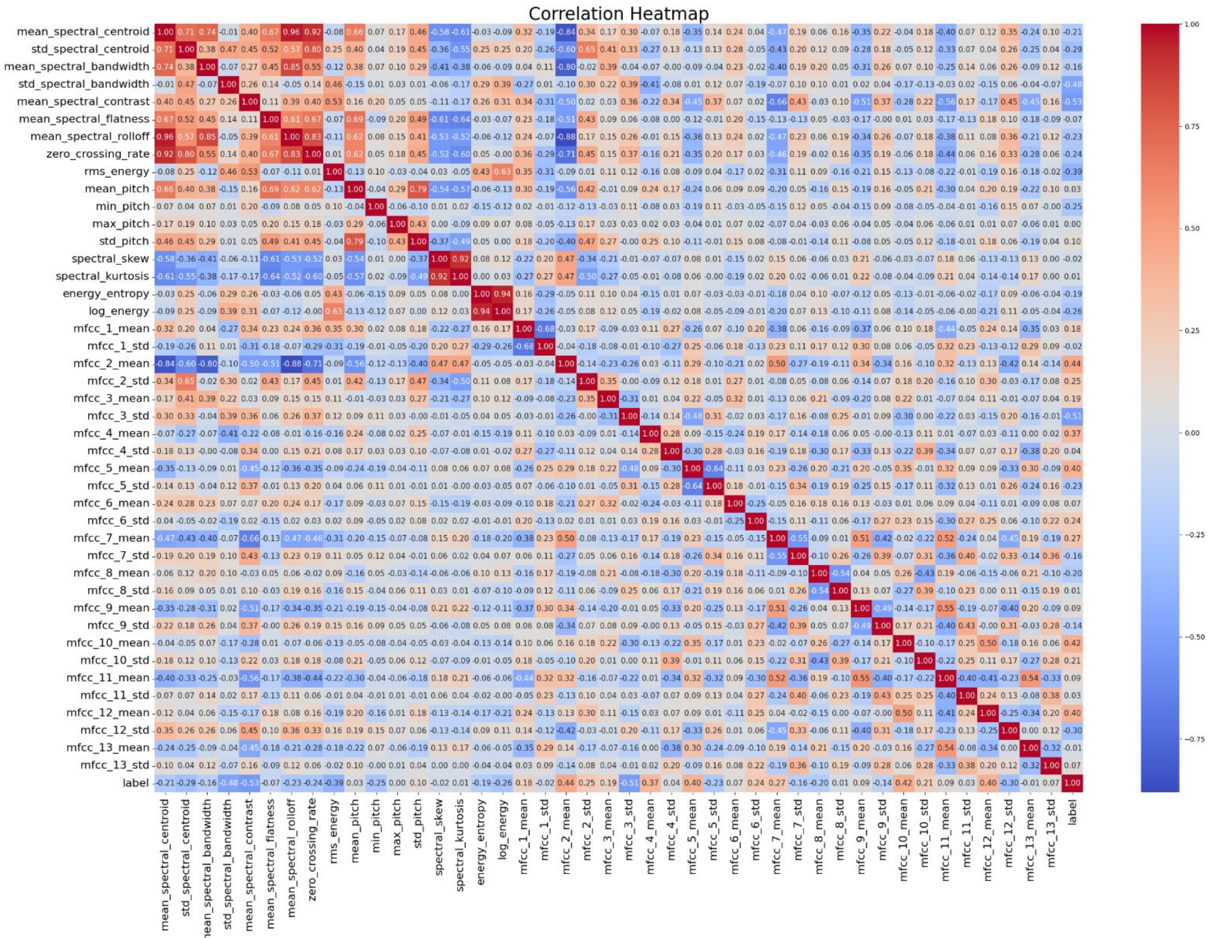

Figure 3: Correlation heatmap

Distribution (Target Value)

The figure 4 bar chart shows the distribution of the target variable (gender) in the dataset. There are more male (label 1) samples than female (label 0) samples. This imbalanced number of target values might affect the accuracy of the model between class 0 and 1. So, accuracy alone might be misleading. It's better to also check precision, recall, and F1-score for both classes.

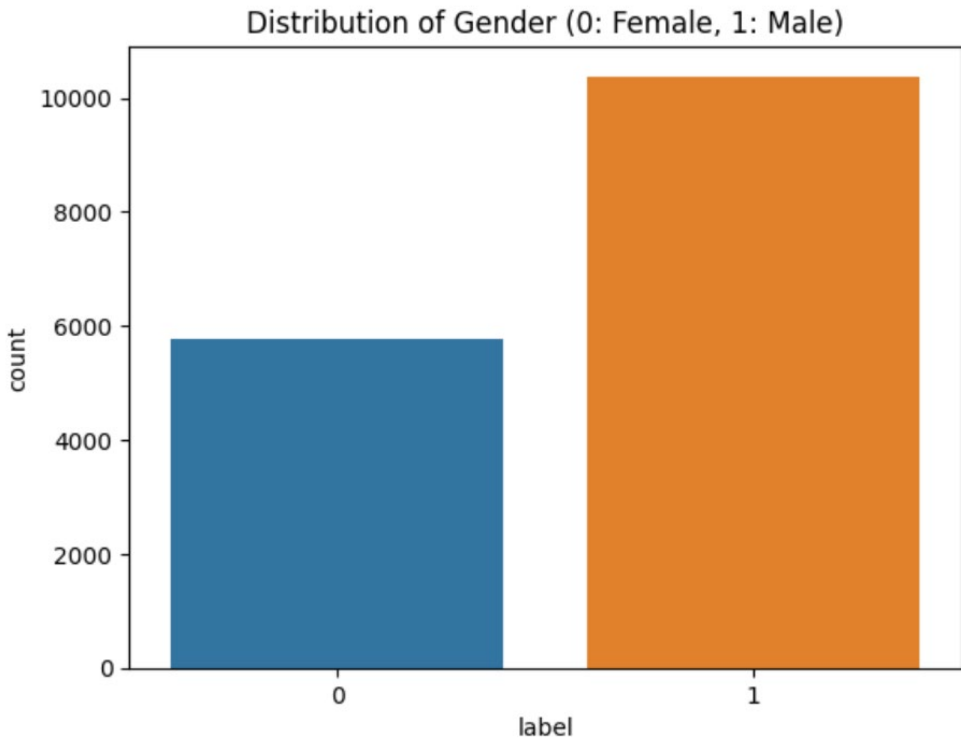

Figure 4: Gender count bar chart

Features:

- **mean_spectral_centroid:** The average spectral centroid, representing the "centre of mass" of the spectrum, indicating brightness.
- **std_spectral_centroid:** The standard deviation of the spectral centroid, measuring variability in brightness.
- **mean_spectral_bandwidth:** The average width of the spectrum, reflecting how spread out the frequencies are.
- **std_spectral_bandwidth:** The standard deviation of spectral bandwidth, indicating variability in frequency spread.
- **mean_spectral_contrast:** The average difference between peaks and valleys in the spectrum, indicating tonal contrast.
- **mean_spectral_flatness:** The average flatness of the spectrum, measuring the noisiness of the signal.
- **mean_spectral_rolloff:** The average frequency below which a specified percentage of the spectral energy resides, indicating sharpness.
- **zero_crossing_rate:** The rate at which the signal crosses the zero-amplitude axis, representing noisiness or percussiveness.
- **rms_energy:** The root means square energy of the signal, reflecting its loudness.
- **mean_pitch:** The average pitch frequency of the audio.

- **min_pitch:** The minimum pitch frequency.
- **max_pitch:** The maximum pitch frequency.
- **std_pitch:** The standard deviation of pitch frequency, measuring variability in pitch.
- **spectral_skew:** The skewness of the spectral distribution, indicating asymmetry.
- **spectral_kurtosis:** The kurtosis of the spectral distribution, indicating the peakiness of the spectrum.
- **energy_entropy:** The entropy of the signal energy, representing its randomness.
- **log_energy:** The logarithmic energy of the signal, a compressed representation of energy.
- **mfcc_1_mean to mfcc_13_mean:** The mean of the first 13 Mel Frequency Cepstral Coefficients (MFCCs), representing the timbral characteristics of the audio.
- **mfcc_1_std to mfcc_13_std:** The standard deviation of the first 13 MFCCs, indicating variability in timbral features.
- **label:** The target variable indicating the gender, male (1) or female (0).

Data Preprocessing

We started by creating a copy of the data to avoid modifying the original dataset. To initialise the features for machine learning models, we used the normalisation technique, the z-score method. This standardizes the feature values so that they have a mean of 0 and a standard deviation of 1. It is important for models that rely on distance-based calculations, such as k-NN or SVM, to ensure all features contribute equally regardless of their original scales. We didn't remove any features or columns; instead, we checked the dataset for duplicate rows. We identified 1078 duplicate data points and removed them to ensure data quality and minimise redundancy. This step helps improve model performance and training speed by eliminating unnecessary repeated information.

	mean_spectral_centroid	std_spectral_centroid	mean_spectral_bandwidth	std_spectral_bandwidth	mean_spectral_contrast
0	1.717415	0.491121	2.226455	-0.775375	0.657270
1	0.281028	-0.075386	1.349508	-0.239354	0.701070
2	0.869924	-0.027844	1.265341	0.435550	0.076009
3	1.058172	1.025873	1.250729	-0.128357	-0.502745
4	0.119538	0.260597	0.299175	0.284340	-0.053971
...	...	...	...	...	...
15065	0.560669	0.259796	1.699631	0.678830	-0.253084
15066	0.343578	0.372531	1.909786	0.832084	-0.744390
15067	0.373818	0.627458	1.449411	1.079861	-0.508676
15068	-0.203733	0.131926	1.272113	0.956754	-0.742381
15069	-0.304195	0.019202	0.919478	1.681350	-0.645461

Figure 5: Data after normalization

Figure 6 shows all rows of the data distribution after scaled features (normalization).

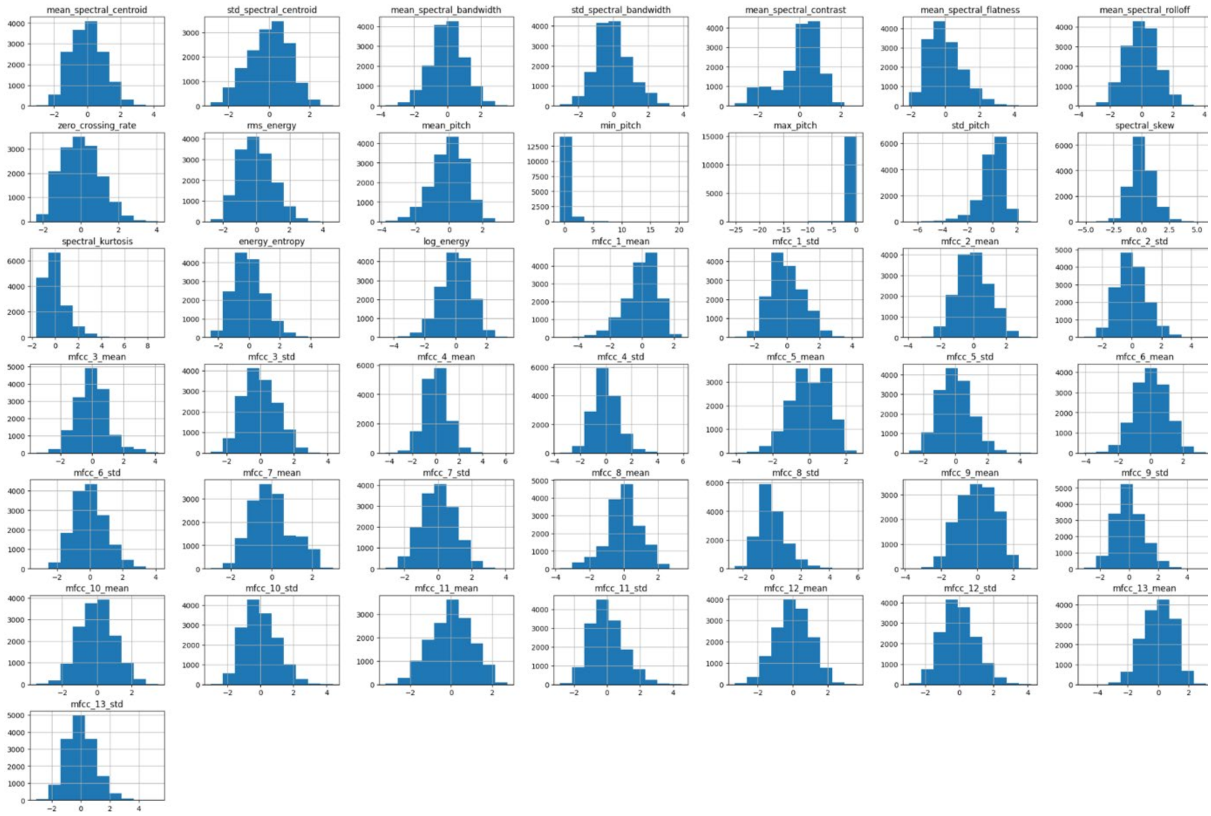

Figure 6: Distribution of Scaled Features

Association Rule Mining (ARM)

Association Rule Mining (ARM) is a data mining technique used to discover interesting relationships or patterns between variables in large datasets. In this project, we applied the Apriori algorithm to identify frequent combinations of binned audio features (such as mfcc_5_mean, mean_spectral_contrast, and mean_pitch) and gender labels. By converting numerical features into categories (e.g., Low, Medium, High), we were able to find meaningful rules — for example, when mean_spectral_contrast is Low, there is a high likelihood the voice belongs to a Male. These rules help us better understand how combinations of audio features are associated with gender, offering insights that support both model interpretation and feature selection.

Data Modelling Classification

After importing the necessary library, we initialized the training data using a train-test split. This step divides our dataset into two parts: 80% for training the model and 20% for testing the model's performance. We also used the normalized features to ensure that all variables are on the same scale, which improves model accuracy. A train-test split allows the machine learning model to learn patterns from the training set and evaluate how well it performs on the testing set. We will visualize the importance of the feature using a random forest classifier, and we will be using two models, Naïve Bayes and Decision Tree, for accuracy training in this report.

Feature Importance (Random Forest Classifier)

We used a Random Forest Classifier to identify the most important features in the dataset. This helps us understand which variables have the most influence on the target label. The bar chart displays the importance of each feature. Higher values mean the feature contributes more to the prediction.

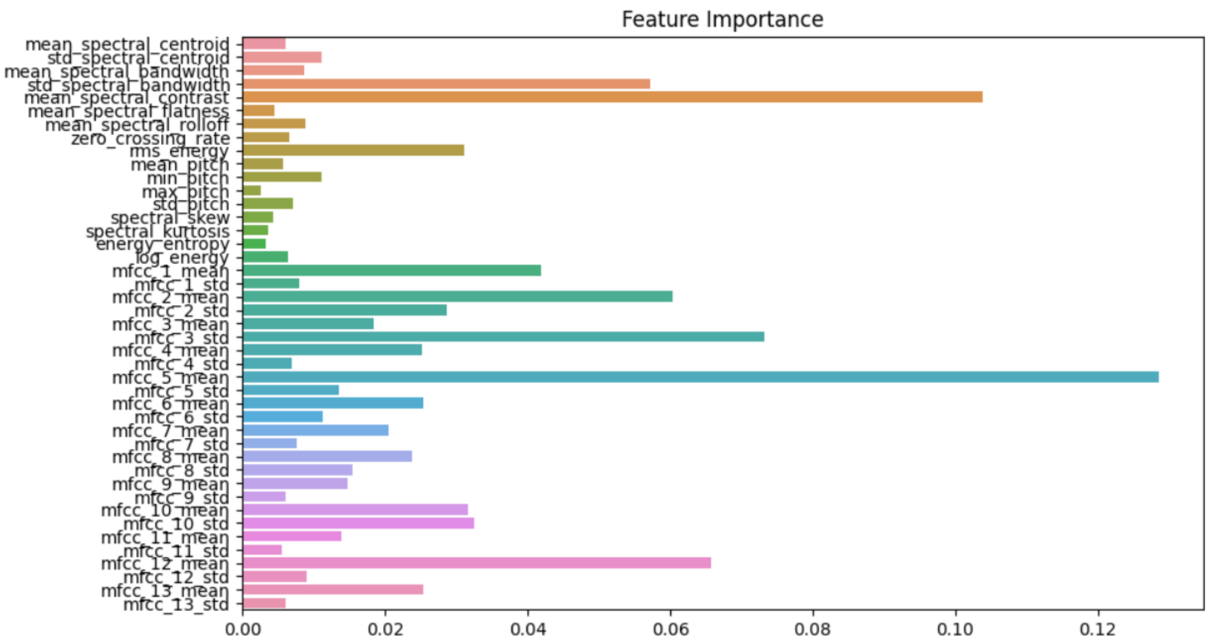

Figure 7: Feature Importance graph

Based on Figure 7, we can observe that not all features contribute equally to the model's prediction of gender. The most significant features, for example, are mfcc_5_mean, mean_spectral_contrast, mfcc_3_std and std_spectral_bandwidth. These features have higher importance scores, indicating that they play a more important role in classifying gender from audio data. On the other hand, several features like mean_spectral_centroid and mfcc_13_std have very low importance and contribute little to the model's decision-making process. This analysis can help with feature selection and dimensionality reduction in future models to improve performance and efficiency.

Decision Tree

The Decision Tree classifier is a rule-based model that splits the data into branches based on feature values, creating a tree-like structure. At each node, the model chooses the feature and threshold that best separate the data using a criterion such as entropy or gini impurity. This allows the model to learn simple decision rules for classification. In our case, we used entropy as the splitting criterion, which helps in building a more informative tree. Decision Trees are intuitive and easy to visualize, and they handle both numerical and categorical data effectively (Raschka & Mirjalili, 2019).

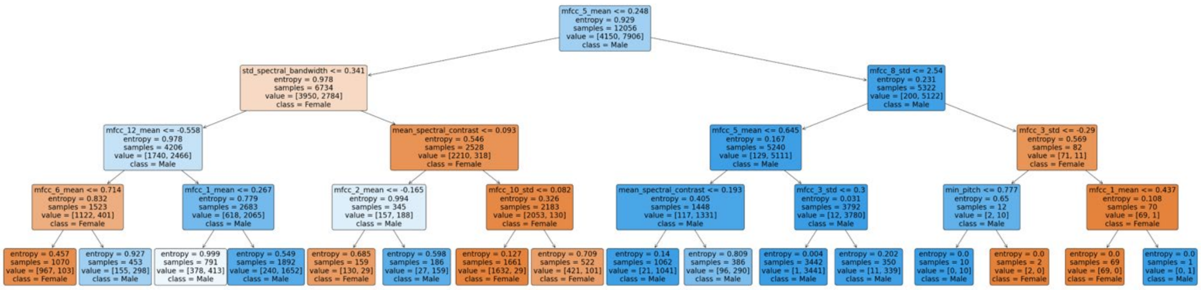

Figure 8: Decision Tree plot

The Decision Tree classifier first splits data into smaller groups based on feature values. For a better understanding of how it works, we visualize the plot of the decision tree shown in Figure 8. After running this model, the result shows a high accuracy of 90% on both the training and testing sets.

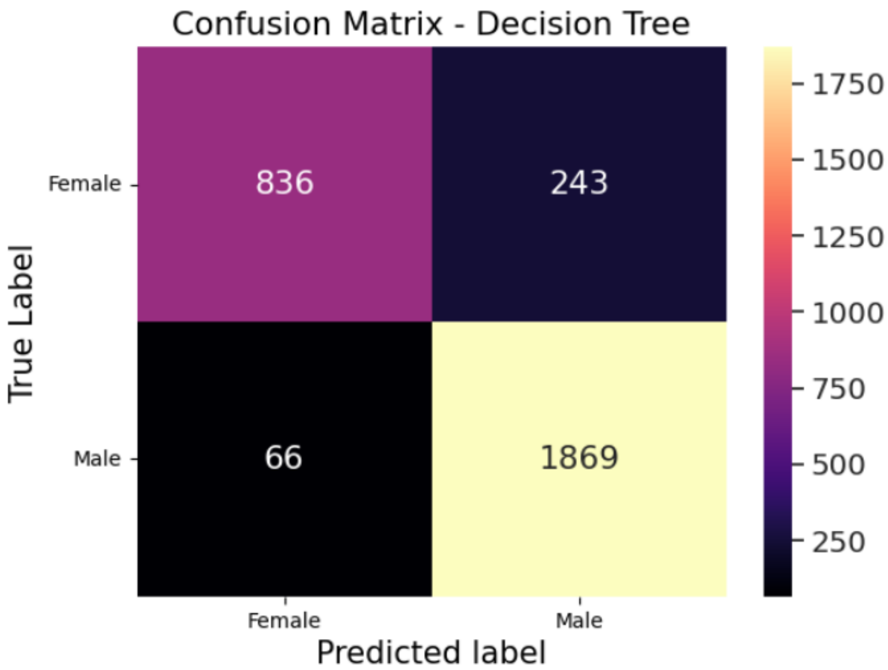

Figure 9: Decision Tree Confusion Matrix

For better observation in the accuracy, we used a confusion matrix plot. From the confusion matrix shown in Figure 9, we can observe that the Decision Tree model performs well in classifying male voices but slightly struggles with female voices.

- True Negative (Female correctly predicted as Female): 836
- False Positive (Female incorrectly predicted as Male): 243
- True Positive (Male correctly predicted as Male): 1869
- False Negative (Male incorrectly predicted as Female): 66

```
Decision Tree Classification Report:
              precision    recall  f1-score   support

           0       0.93      0.77      0.84      1079
           1       0.88      0.97      0.92      1935

    accuracy                           0.90      3014
   macro avg       0.91      0.87      0.88      3014
weighted avg       0.90      0.90      0.90      3014
```

Figure 10: Classification Report (Decision Tree)

In order to go deeper and study the result of the model, we visualise it using the classification report shown in Figure 10. The Decision Tree model achieved a high overall accuracy of 90%, showing strong performance in gender classification. However, the performance varies across classes:

- Female (Label 0): Precision is high (0.93), but recall is lower (0.77), meaning the model correctly identifies most predicted females but misses many actual female samples.
- Male (Label 1): Precision and recall are both high (0.88 and 0.97), indicating the model is very effective at identifying male voices.

In conclusion, this may be caused by the imbalance of the target variable (gender) in the dataset, resulting in better performance on male predictions. While the model is generally reliable in high accuracy performance, its recall for female samples could be improved for more balanced classification.

Naïve Bayes (GaussianNB)

The Naïve Bayes classifier is a probabilistic model based on Bayes' Theorem, assuming that all features are independent of each other. We used the Gaussian Naïve Bayes

(GaussianNB) variant, which assumes that the features follow a normal distribution, making it suitable for continuous data such as audio features (Ibm, n.d..).

After training the model using the training dataset, the GaussianNB model achieved an impressive accuracy of 93% on both the training and testing sets, indicating that it generalizes well without overfitting. This high performance suggests that GaussianNB is effective for this classification task, particularly when feature scaling is properly handled. Its simplicity and speed make it a strong baseline model in binary classification problems such as gender recognition from voice features.

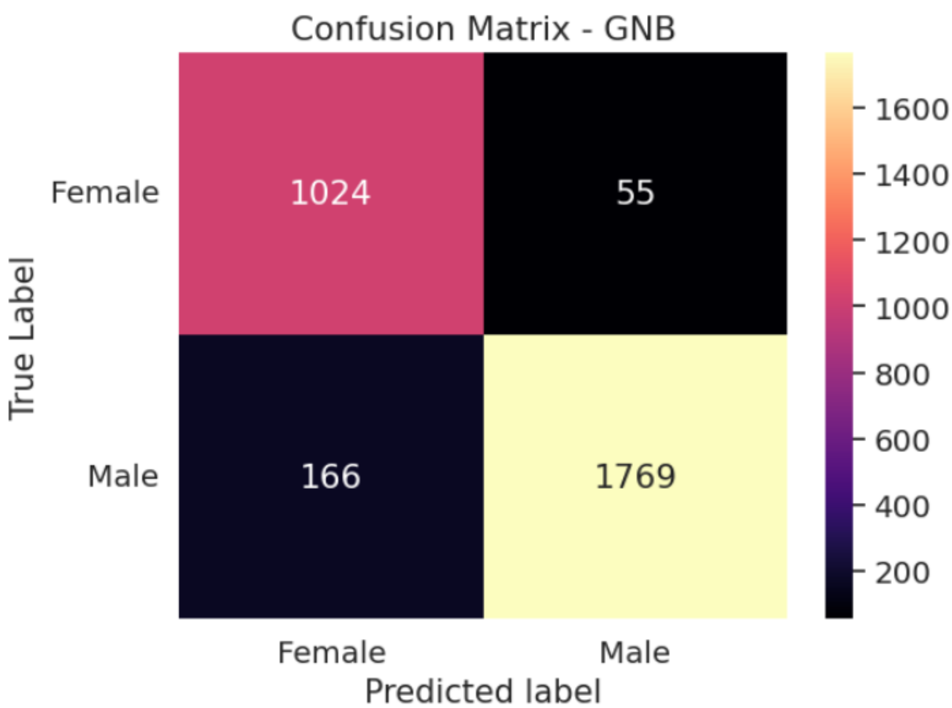

Figure 11: Naïve Bayes Confusion Matrix

We use the confusion matrix technique for this model also to visualise the accuracy. The confusion matrix shown in Figure 11 demonstrates the performance of the GaussianNB model in classifying gender based on voice features. The matrix shows the number of correct and incorrect predictions for each class:

- True Negative (Female correctly predicted as Female): 1024
- False Positive (Female incorrectly predicted as Male): 55
- True Positive (Male correctly predicted as Male): 1769
- False Negative (Male incorrectly predicted as Female): 166

Naive Bayes Classification Report:

	precision	recall	f1-score	support
0	0.86	0.95	0.90	1079
1	0.97	0.91	0.94	1935
accuracy			0.93	3014
macro avg	0.92	0.93	0.92	3014
weighted avg	0.93	0.93	0.93	3014

Figure 12: Classification Report (Naïve Bayes)

The classification report shown in Figure 12 further supports the effectiveness of the GaussianNB model in gender classification based on voice features. The model achieved an impressive overall accuracy of 93%, indicating that it performs well on unseen data.

- Female (Label 0): Precision is moderate (0.86), but recall is high (0.95), meaning the model correctly identifies most actual female samples, though some predictions for females may include a few incorrect ones.
- Male (Label 1): Precision is very high (0.97), and recall is also strong (0.91), indicating the model is highly reliable when predicting male voices and correctly captures most of the actual male samples.

Overall, the GaussianNB model shows strong and balanced performance even if the target variable was slightly imbalanced, especially with high recall for female samples and excellent precision for male samples. This makes it a strong candidate for gender classification in voice datasets.

Discussion and Comparisons

The Decision Tree Model achieved a strong result with balanced precision and recall for male samples, but it shows slightly lower performance on female samples due to the lower recall (0.77). This means that during the actual training section, this model fails to classify quite a few female samples. However, the decision tree is easy to interpret and visualise, which makes it useful for understanding the decision-making processes.

On the other hand, the Naïve Bayes Model (GaussianNB) outperformed the Decision Tree slightly in the result of accuracy (93% vs 90%) and more balance on recall for females (0.95 vs 0.77). This model is much better at identifying actual female voice, however, it had more false positives when predicting male samples, but overall, it seems to be more balanced between males and females on recall and predicting within this model result. Naïve Bayes also benefits from its simplicity and efficiency, especially with continuous input data that fits a normal distribution.

The imbalance of the target variable in the dataset (about 50% more male samples than female) may explain why both models perform slightly better on male predictions. However, the result shows that Naïve Bayes appears to be more resilient to this imbalance.

Conclusion

In conclusion, based on the result and performance analysis, the GaussianNB model is the most suitable method for this classification task. It achieved the highest accuracy on both training and testing datasets and showed a more balanced performance across precision and recall gender classes. Its simplicity, speed and ability to handle continuous input features make it a strong model for real-world applications such as gender recognition in voice-based virtual assistants, customer service systems or biometric identity verification.

In real-world usage, the GaussianNB model can be used in a voice-based system to personalize user experience, for example, by adjusting voice tones, language, or interaction style based on the detected gender. Its lightweight advantages also make it suitable to implement on devices with limited computational resources.

One of the methods and then explain why and how it can be applied to the real world

Possible future works and improvements

Although the models performed well at percentages above 75, there are still several ways to enhance the system in the future. The main issue that the Decision Tree to be inaccurate at the female recall part is cause by the imbalance of target variable in this dataset. This can be improved through techniques like stratified sampling or SMOTE, also known as the Synthetic Minority Over-Sampling technique. Moreover, exploring with more advanced models such as Support Vector Machine (SVM), Random Forest, or even deep learning models like Neural Network may result in a higher performance, especially when dealing with a complex or large dataset.

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